

Climate Change: Causes, Effects, and Mitigation Strategies

Introduction

Climate change refers to long-term shifts in global or regional climate patterns. While climate has changed throughout Earth's history, the current rapid warming is primarily attributed to human activities, particularly the burning of fossil fuels and deforestation.

The Greenhouse Effect

The greenhouse effect is a natural phenomenon where greenhouse gases in Earth's atmosphere (including water vapor, carbon dioxide, and methane) trap some of the sun's heat near the surface, making the planet habitable. However, human activities have significantly increased the concentration of these gases, intensifying the greenhouse effect and causing global warming.

Primary Causes of Climate Change

The primary drivers of climate change are greenhouse gas emissions, particularly carbon dioxide from burning fossil fuels like coal, oil, and natural gas. Deforestation contributes by reducing the planet's capacity to absorb CO₂. Industrial processes and agriculture also release significant amounts of methane and nitrous oxide.

Global Temperature Rise

Since the pre-industrial era, the average global temperature has risen by approximately 1.1 degrees Celsius. The last decade was the warmest on record. Scientists project that without significant emission reductions, temperatures could rise by 2.5 to 4.5 degrees Celsius by the end of this century.

Sea Level Rise

Rising sea levels pose severe threats to coastal regions worldwide. Effects include increased coastal flooding, accelerated erosion, loss of coastal wetlands and habitats, saltwater contamination of freshwater resources, and the potential displacement of millions of people living in low-lying areas. Current projections suggest sea levels could rise by 0.3 to 1.0 meters by 2100.

Extreme Weather Events

Climate change is increasing the frequency and intensity of extreme weather events, including hurricanes, droughts, heatwaves, and heavy rainfall. These events cause significant economic damage, displace communities, and threaten food and water security worldwide.

The Paris Agreement

The Paris Agreement, adopted in December 2015, is a landmark international accord within the United Nations Framework Convention on Climate Change. Its central aim is to strengthen the global response to climate change by keeping the global temperature rise this century well below 2 degrees Celsius above pre-industrial levels, while pursuing efforts to limit the increase to 1.5 degrees Celsius.

Renewable Energy Solutions

Renewable energy sources including solar photovoltaic, wind turbines, and hydroelectric power generate electricity without directly emitting greenhouse gases during operation. Transitioning from fossil fuels to renewables is one of the most effective strategies for reducing carbon emissions and mitigating climate change.

Carbon Capture and Storage

Carbon capture and storage (CCS) is a technology that captures CO₂ emissions from sources like power plants and industrial processes, and stores it underground to prevent it from entering the atmosphere. While promising, the technology is still expensive and not yet deployed at scale.

Individual Actions

Individuals can contribute to climate change mitigation by reducing energy consumption, using public transportation, eating less meat, reducing waste, and supporting renewable energy. While individual actions alone are insufficient, they contribute to broader societal change and market signals.